Quantum Mechanics 3

Worked Solutions

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Introduction

This document presents worked solutions to the *Quantum Mechanics 3* problem sheet supplied with the BPhO Computational Physics Challenge 2026. The sheet focuses on de Broglie matter waves, the uncertainty principle, wavefunctions, the particle in a box, and the Copenhagen interpretation of measurement.

The derivations and numerical methods follow the accompanying BPhO solution document. Numerical values are quoted to a sensible precision and use the physical constants supplied on the problem sheet.

1 Question 1 — Matter waves, uncertainty and measurement

1.1 (i) de Broglie wavelength of an electron

For a non-relativistic particle,

$$p = m_e v,$$

and the de Broglie relation is

$$\lambda = \frac{h}{p}.$$

The electron travels at

$$v = \frac{c}{137}.$$

Therefore

$$\lambda = \frac{h}{m_e c/137} = \frac{137h}{m_e c}.$$

Using

$$h = 6.626 \times 10^{-34}\ \mathrm{J\,s}, \qquad m_e = 9.109 \times 10^{-31}\ \mathrm{kg}, \qquad c = 2.998 \times 10^{8}\ \mathrm{m\,s^{-1}},$$

gives

$$\lambda = \frac{137(6.626 \times 10^{-34})}{(9.109 \times 10^{-31})(2.998 \times 10^{8})} \approx 3.3 \times 10^{-10}\ \mathrm{m}.$$

Thus

$$\boxed{\lambda \approx 3.3 \times 10^{-10}\ \mathrm{m}}.$$

The supplied solution also checks that using the relativistic momentum makes no significant difference at this speed to the quoted precision.

1.2 (ii) Equal de Broglie wavelengths for an electron and neutron

If

$$\lambda_e = \lambda_n,$$

then

$$\frac{h}{p_e} = \frac{h}{p_n},$$

so

$$p_e = p_n.$$

Using classical momentum,

$$m_e v_e = m_n v_n.$$

Therefore

$$\boxed{\frac{v_e}{v_n} = \frac{m_n}{m_e}}.$$

Numerically,

$$\frac{v_e}{v_n} = \frac{1.6749 \times 10^{-27}}{9.109 \times 10^{-31}} \approx 1837.$$

Hence

$$\boxed{\frac{v_e}{v_n} \approx 1837}.$$

The classical kinetic energy is

$$E = \frac{p^2}{2m}.$$

Since the momenta are equal,

$$\frac{E_e}{E_n} = \frac{m_n}{m_e} \approx 1837.$$

Thus

$$\boxed{\frac{E_e}{E_n} \approx 1837}.$$

So for equal de Broglie wavelength, the electron travels about 1837 times faster and has about 1837 times the kinetic energy of the neutron.

1.3 (iii) Range and lifetime of a Z boson

The problem gives

$$\Delta x\,\Delta p \approx \frac{\hbar}{2}, \qquad \Delta E\,\Delta t \approx \frac{\hbar}{2},$$

with

$$\Delta p \approx mc, \qquad \Delta E \approx mc^2.$$

Hence

$$\Delta x \approx \frac{\hbar}{2mc},$$

and

$$\Delta t \approx \frac{\hbar}{2mc^2}.$$

The mass of the Z boson is

$$m = \frac{91.2 \text{ GeV}}{c^2}.$$

Converting this to kilograms,

$$m = \frac{91.2 \times 10^9 (1.602 \times 10^{-19})}{(2.998 \times 10^8)^2} \approx 1.626 \times 10^{-25} \text{ kg.}$$

Therefore

$$\Delta x \approx \frac{6.626 \times 10^{-34}}{2(1.626 \times 10^{-25})(2.998 \times 10^8)} \approx 1.08 \times 10^{-17} \text{ m.}$$

Thus

$$\boxed{\Delta x \approx 1.08 \times 10^{-17} \text{ m}}.$$

Similarly,

$$\Delta t \approx \frac{6.626 \times 10^{-34}}{2(1.626 \times 10^{-25})(2.998 \times 10^8)^2} \approx 3.61 \times 10^{-27} \text{ s.}$$

Hence

$$\boxed{\Delta t \approx 3.61 \times 10^{-27} \text{ s}}.$$

These scales are far below ordinary atomic dimensions and timescales.

1.4 (iv) Verifying a solution of the wave equation

Consider

$$\psi(x,t) = A \sin(kx - \omega t).$$

The spatial derivatives are

$$\frac{\partial \psi}{\partial x} = kA \cos(kx - \omega t),$$

and therefore

$$\frac{\partial^2 \psi}{\partial x^2} = -k^2 A \sin(kx - \omega t) = -k^2 \psi.$$

The time derivatives are

$$\frac{\partial \psi}{\partial t} = -\omega A \cos(kx - \omega t),$$

and

$$\frac{\partial^2 \psi}{\partial t^2} = -\omega^2 A \sin(kx - \omega t) = -\omega^2 \psi.$$

Now

$$k = \frac{2\pi}{\lambda}, \qquad \omega = 2\pi f, \qquad c = f\lambda.$$

Thus

$$\omega = ck,$$

so

$$\omega^2 = c^2 k^2.$$

Therefore

$$\frac{\partial^2 \psi}{\partial t^2} = -c^2 k^2 \psi = c^2 \frac{\partial^2 \psi}{\partial x^2}.$$

Hence

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}},$$

which is precisely the one-dimensional wave equation.

1.5 (v) Wavefunction collapse and the double-slit experiment

In an electron double-slit experiment, electrons are detected individually on a screen, but after many electrons have passed through the apparatus, an interference pattern develops. The electrons therefore exhibit wave-like behaviour.

The de Broglie wavelength of the electrons determines the characteristic diffraction scale. If the electrons are accelerated through a voltage V, their classical kinetic energy is

$$E = eV,$$

and, using

$$E = \frac{p^2}{2m},$$

their wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}.$$

The double slit produces an interference pattern because the wavefunction contains contributions associated with both possible paths.

If the electron is instead observed in a way that determines which slit it passed through, the two-path superposition is destroyed. The wavefunction collapses onto the corresponding measurement eigenstate, and the interference pattern disappears.

Thus the act of measurement changes the state being measured: before the which-path measurement, the electron is described by a superposition of the possible paths; after the measurement, it has a definite measured path.

1.6 (vi) Classical kinetic energy in terms of momentum

Classically,

$$E = \frac{1}{2}mv^2$$

and

$$p = mv.$$

Therefore

$$v = \frac{p}{m}.$$

Substituting,

$$E = \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{1}{2}m\frac{p^2}{m^2}.$$

Hence

$$\boxed{E = \frac{p^2}{2m}}.$$

For comparison, the relativistic energy–momentum relation is

$$E^2 = p^2c^2 + m^2c^4,$$

or

$$\boxed{E = \sqrt{p^2c^2 + m^2c^4}},$$

which reduces to $E = pc$ for a photon.

1.7 (vii) Neutron diffraction

The atomic lattice spacing is

$$s = 2.15 \times 10^{-10} \text{ m}.$$

We interpret the stated 30° angular spacing as the angle of the first diffraction maximum from the central maximum, consistent with the supplied solution. Thus

$$\theta_1 = 30^\circ.$$

For first-order Bragg diffraction,

$$s \sin\theta_1 = n\lambda,$$

with $n = 1$. Therefore

$$\lambda = s \sin 30^\circ = \frac{s}{2},$$

so

$$\boxed{\lambda = \frac{s}{2}}.$$

Thus

$$\lambda = \frac{2.15 \times 10^{-10}}{2} = 1.08 \times 10^{-10} \text{ m}.$$

Hence

$$\boxed{\lambda \approx 1.08 \times 10^{-10} \text{ m}}.$$

Using

$$p = \frac{h}{\lambda}$$

and

$$E = \frac{p^2}{2m_n},$$

we obtain

$$E = \frac{h^2}{2m_n\lambda^2}.$$

Numerically,

$$E \approx 0.071 \text{ eV}.$$

Therefore

$$\boxed{E \approx 0.071 \text{ eV}}.$$

The neutron speed follows from

$$p = m_n v,$$

so

$$v = \frac{h}{m_n\lambda}.$$

Hence

$$\boxed{v \approx 3.68 \times 10^3 \text{ m s}^{-1}}.$$

The supplied solution also compares this energy with the thermal energy scale

$$\frac{3}{2}k_B T$$

at room temperature and notes that the neutron energy is of the same order.

1.8 (viii) Photoelectric effect and an equal de Broglie wavelength

For silver,

$$\phi = 4.3 \text{ eV},$$

and the maximum photoelectron kinetic energy is

$$K_{\max} = 5.1 \text{ eV}.$$

The photoelectric equation is

$$K_{\max} = \frac{hc}{\lambda} - \phi.$$

Therefore

$$\lambda = \frac{hc}{K_{\max} + \phi}.$$

Using

$$hc \approx 1240 \text{ eV nm},$$

we find

$$\lambda = \frac{1240}{5.1 + 4.3} \approx 131.9 \text{ nm}.$$

Thus

$$\boxed{\lambda \approx 131.9 \text{ nm}}.$$

Now consider an electron with the same de Broglie wavelength. Its kinetic energy is

$$E_e = \frac{p^2}{2m_e},$$

with

$$p = \frac{h}{\lambda}.$$

Hence

$$E_e = \frac{h^2}{2m_e \lambda^2}.$$

Substitution gives

$$\boxed{E_e \approx 8.65 \times 10^{-5} \text{ eV}}.$$

Using the wavelength above gives

$$E_e \approx 8.65 \times 10^{-5} \text{ eV},$$

consistent with the supplied handwritten solution. This illustrates the enormous difference between the energy of a UV photon and the kinetic energy of a non-relativistic electron having the same wavelength.

1.9 (ix) Electron microscope resolution

The resolution is approximately the electron wavelength:

$$\lambda \approx \Delta x.$$

Given

$$\Delta x = 0.42 \text{ nm},$$

we therefore require

$$\lambda \approx 0.42 \text{ nm}.$$

Using

$$E = \frac{h^2}{2m_e\lambda^2},$$

we obtain

$$E = \frac{(6.626 \times 10^{-34})^2}{2(9.109 \times 10^{-31})(0.42 \times 10^{-9})^2}.$$

Converting to electron-volts,

$$\boxed{E \approx 8.53 \text{ eV}}.$$

Equivalently, if the electron is accelerated through a voltage V, then

$$E = eV,$$

and the non-relativistic relation becomes

$$\boxed{\frac{\lambda}{\text{nm}} \approx \frac{1.23}{\sqrt{V}}},$$

where V is measured in volts.

For larger accelerating voltages, relativistic corrections become important. The supplied solution gives

$$\boxed{\frac{\lambda}{\text{nm}} \approx \frac{1.23}{\sqrt{V}\sqrt{1 + 9.78 \times 10^{-7}V}}}$$

as the corresponding relativistic expression.

1.10 (x) Time and spatial extent of beta decay

The emitted electron has energy

$$\Delta E = 5 \text{ MeV}.$$

From the energy-time uncertainty principle,

$$\Delta E\,\Delta t \gtrsim \frac{\hbar}{2},$$

so

$$\Delta t \gtrsim \frac{\hbar}{2\Delta E}.$$

Therefore

$$\boxed{\Delta t \gtrsim 6.58 \times 10^{-23} \text{ s}}.$$

For the spatial extent, the problem assumes

$$\Delta p \approx \sqrt{2m_e\Delta E}.$$

Thus

$$\Delta x \gtrsim \frac{\hbar}{2\Delta p} = \frac{\hbar}{2\sqrt{2m_e\Delta E}}.$$

Substitution gives

$$\boxed{\Delta x \gtrsim 4.36 \times 10^{-14} \text{ m}}.$$

This is a nuclear-scale distance, consistent with the short-range nature of the process.

2 Question 2 — Particle in an infinite potential box

A particle of mass m is confined to

$$0 \le x \le a$$

with

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x \le 0 \text{ or } x \ge a. \end{cases}$$

2.1 (i) Standing-wave solutions

The infinite potential walls mean that the particle cannot exist outside the interval $0 < x < a$. Therefore the wavefunction must vanish at the walls:

$$\psi(0,t) = 0, \qquad \psi(a,t) = 0.$$

A standing wave satisfying the first boundary condition can be written as

$$\psi(x,t) = A_n \sin\left(\frac{n\pi x}{a}\right) \cos \omega t.$$

At $x = a$,

$$\sin(n\pi) = 0$$

only when

$$\boxed{n = 1, 2, 3, \ldots}.$$

Thus n must be a positive integer.

The wavelength is related to the box length by

$$\frac{n\lambda_n}{2} = a,$$

so

$$\lambda_n = \frac{2a}{n}.$$

Therefore

$$k_n = \frac{2\pi}{\lambda_n} = \frac{n\pi}{a}.$$

The standing-wave picture is natural because the infinitely high walls impose fixed nodes at the two ends of the box.

2.2 (ii) Energy eigenvalues

Inside the box,

$$V = 0,$$

so the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} = E\psi.$$

For

$$\psi = A_n \sin\left(\frac{n\pi x}{a}\right),$$

we have

$$\frac{\partial^2 \psi}{\partial x^2} = -\left(\frac{n\pi}{a}\right)^2 \psi.$$

Therefore

$$\frac{\hbar^2}{2m}\left(\frac{n\pi}{a}\right)^2\psi = E\psi.$$

Hence

$$\boxed{E_n = \frac{\hbar^2\pi^2 n^2}{2ma^2}}.$$

For a proton,

$$m = m_p = 1.6726 \times 10^{-27}\ \text{kg},$$

and

$$a = 0.47 \times 10^{-15}\ \text{m}.$$

Substitution gives

$$E_n \approx 927.2\, n^2\ \text{MeV}.$$

Thus for the ground state,

$$\boxed{E_1 \approx 927.2\ \text{MeV}}.$$

The proton rest-mass energy is

$$m_p c^2 \approx 938.4\ \text{MeV}.$$

Therefore

$$\boxed{\frac{E_1}{m_p c^2} \approx 0.988}.$$

The ground-state confinement energy is therefore almost equal to the proton rest-mass energy. The non-relativistic Schrödinger result is consequently outside its regime of validity, but the numerical comparison is exactly what the question asks for.

2.3 (iii) Time dependence and normalisation

The factor $\cos\omega t$ is a real representation of a standing-wave oscillation, but the Schrödinger wavefunction is generally complex. For a stationary energy eigenstate, the correct time dependence is

$$e^{-iEt/\hbar}.$$

Thus

$$\psi_n(x,t) = A_n e^{-iE_n t/\hbar}\sin\left(\frac{n\pi x}{a}\right).$$

The Born interpretation requires

$$\int_{-\infty}^{\infty} |\psi|^2\, dx = 1.$$

Since the wavefunction is zero outside the box,

$$\int_0^a |\psi|^2\, dx = 1.$$

The phase factors cancel:

$$|\psi|^2 = A_n^2 \sin^2\left(\frac{n\pi x}{a}\right).$$

Hence

$$A_n^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1.$$

Using

$$\int_0^a \sin^2\left(\frac{n\pi x}{a}\right)\,dx = \frac{a}{2},$$

we find

$$A_n^2 \frac{a}{2} = 1,$$

so

$$\boxed{A_n = \sqrt{\frac{2}{a}}}.$$

Therefore the normalised stationary state is

$$\boxed{\psi_n(x,t) = \sqrt{\frac{2}{a}}\, e^{-iE_n t/\hbar} \sin\left(\frac{n\pi x}{a}\right)}$$

for $0 < x < a$.

2.4 (iv) Momentum expectation values

A stationary standing wave is a superposition of waves travelling in opposite directions. Consequently the average momentum is zero:

$$\boxed{\langle p \rangle = 0}.$$

The energy is

$$E = \frac{p^2}{2m},$$

so

$$p^2 = 2mE.$$

Using

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2ma^2},$$

we obtain

$$\langle p^2 \rangle = 2mE_n = \frac{\hbar^2 \pi^2 n^2}{a^2}.$$

Thus

$$\boxed{\langle p^2 \rangle = \frac{\hbar^2 \pi^2 n^2}{a^2}}.$$

Since $\langle p \rangle = 0$,

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \frac{\hbar n \pi}{a}.$$

2.5 (v) Position expectation values

The expectation value of position is

$$\langle x \rangle = \int_0^a x|\psi|^2\,dx.$$

Using the normalised wavefunction,

$$\langle x \rangle = \frac{2}{a}\int_0^a x \sin^2\left(\frac{n\pi x}{a}\right)\,dx.$$

Using
$$\int x\sin^2(\alpha x)\,dx = \frac{x^2}{4} - \frac{x\sin(2\alpha x)}{4\alpha} - \frac{\cos(2\alpha x)}{8\alpha^2} + C,$$
with
$$\alpha = \frac{n\pi}{a},$$
and the identities
$$\sin(2n\pi) = 0, \qquad \cos(2n\pi) = 1,$$
we obtain
$$\boxed{\langle x\rangle = \frac{a}{2}}.$$

For the second moment,
$$\langle x^2\rangle = \frac{2}{a}\int_0^a x^2\sin^2\left(\frac{n\pi x}{a}\right)\,dx.$$

Using the corresponding integral supplied on the problem sheet gives
$$\boxed{\langle x^2\rangle = \frac{a^2}{3}\left(1 - \frac{3}{2n^2\pi^2}\right)}.$$

Therefore
$$(\Delta x)^2 = \langle x^2\rangle - \langle x\rangle^2.$$

Hence
$$(\Delta x)^2 = \frac{a^2}{3}\left(1 - \frac{3}{2n^2\pi^2}\right) - \frac{a^2}{4}.$$

Simplifying,
$$(\Delta x)^2 = \frac{a^2}{12} - \frac{a^2}{2n^2\pi^2}.$$

Thus
$$\boxed{\Delta x = \frac{a}{2\pi}\sqrt{\frac{n^2\pi^2}{3} - 2}}.$$

2.6 (vi) Verification of the uncertainty principle

From part (iv),
$$\Delta p = \frac{\hbar n\pi}{a}.$$

Combining this with
$$\Delta x = \frac{a}{2\pi}\sqrt{\frac{n^2\pi^2}{3} - 2},$$
we obtain
$$\Delta x\,\Delta p = \frac{\hbar}{2}\sqrt{\frac{n^2\pi^2}{3} - 2}.$$

Therefore
$$\boxed{\Delta x\,\Delta p = \frac{\hbar}{2}\sqrt{\frac{n^2\pi^2}{3} - 2}}.$$

The smallest possible value occurs for the smallest allowed quantum number,
$$n = 1.$$

Then

$$\sqrt{\frac{\pi^2}{3} - 2} \approx 1.14.$$

Hence

$$\Delta x\,\Delta p \approx 1.14\frac{\hbar}{2} > \frac{\hbar}{2}.$$

Therefore every allowed particle-in-a-box state satisfies

$$\boxed{\Delta x\,\Delta p \geq \frac{\hbar}{2}},$$

which verifies the Heisenberg uncertainty principle.

3 Question 3 — Entangled photons and quantum cryptography

Two vertically polarised entangled photons travel in opposite directions to detectors A and B. The detectors are oriented at angles θ and ϕ.

The problem asks first for a classical calculation in which the measurements are treated as independent, and then for the Copenhagen interpretation in which the first measurement affects the state of the second photon.

3.1 Classical calculation

Using Malus' law, the probabilities for detector A are

$$P(X_A) = \cos^2\theta, \qquad P(Y_A) = \sin^2\theta.$$

Similarly,

$$P(X_B) = \cos^2\phi, \qquad P(Y_B) = \sin^2\phi.$$

The two detectors give different outcomes in the two cases

$$X_A, Y_B$$

and

$$Y_A, X_B.$$

Assuming the measurements are independent,

$$P_{\text{classical}}(\text{mismatch}) = P(X_A)P(Y_B) + P(Y_A)P(X_B).$$

Therefore

$$\boxed{P_{\text{classical}} = \cos^2\theta\sin^2\phi + \sin^2\theta\cos^2\phi}.$$

For

$$\theta = -45^\circ, \qquad \phi = 30^\circ,$$

we have

$$\cos^2(-45^\circ) = \sin^2(-45^\circ) = \frac{1}{2},$$

and

$$\sin^2(30^\circ) = \frac{1}{4}, \qquad \cos^2(30^\circ) = \frac{3}{4}.$$

Hence

$$P_{\text{classical}} = \frac{1}{2}\frac{1}{4} + \frac{1}{2}\frac{3}{4} = \frac{1}{2}.$$

Thus

$$\boxed{P_{\text{classical}}(\text{mismatch}) = \frac{1}{2}}.$$

3.2 Copenhagen interpretation

Suppose detector A measures first.

If A measures X_A, the entangled state collapses so that the corresponding polarisation of the photon arriving at B is referenced to the A measurement axis. The angle between the A and B measurement axes is

$$\phi - \theta.$$

Thus the conditional probabilities for detector B involve

$$\cos^2(\phi - \theta)$$

and

$$\sin^2(\phi - \theta).$$

There are two possibilities for the first measurement:

$$P(X_A) = \cos^2\theta, \qquad P(Y_A) = \sin^2\theta.$$

The probability of a mismatch is therefore

$$P_{\text{quantum}}(\text{mismatch}) = \cos^2\theta\sin^2(\phi - \theta) + \sin^2\theta\cos^2(\phi - \theta).$$

This simplifies to

$$\boxed{P_{\text{quantum}}(\text{mismatch}) = \sin^2(\phi - \theta)}.$$

For

$$\theta = -45^\circ, \qquad \phi = 30^\circ,$$

the relative angle is

$$\phi - \theta = 30^\circ - (-45^\circ) = 75^\circ.$$

Therefore

$$P_{\text{quantum}} = \sin^2(75^\circ).$$

Using

$$\sin 75^\circ = \sin(30^\circ + 45^\circ) = \frac{1}{2\sqrt{2}}(1 + \sqrt{3}),$$

we obtain

$$P_{\text{quantum}} = \frac{(1 + \sqrt{3})^2}{8} \approx 0.933.$$

Hence

$$\boxed{P_{\text{quantum}}(\text{mismatch}) \approx 0.933}.$$

The ratio of the quantum prediction to the classical prediction is

$$\frac{P_{\text{quantum}}}{P_{\text{classical}}} = \frac{(1 + \sqrt{3})^2/8}{1/2} = \frac{(1 + \sqrt{3})^2}{4} \approx 1.87.$$

Thus

$$\boxed{\frac{P_{\text{quantum}}}{P_{\text{classical}}} \approx 1.87}.$$

The quantum result is therefore substantially different from the independent classical prediction. This difference is the basis of the quantum-cryptographic idea described in the problem sheet: the statistics of polarisation measurements can reveal whether an eavesdropper has disturbed the transmitted quantum states.